

LANAI, HI, USA

WMO: 911905

Lat: 20.790N

Lon: 156.949W

Elev: 1300

StdP: 14.02

Time Zone: -10.00 (W10)

Period: 05-19

WBAN: 22524

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	58.6	59.8	51.3	58.9	63.8	53.0	62.8	64.7	36.0	68.0	32.9	66.1	5.0	80	0.382

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.9	83.8	71.6	82.6	71.3	81.8	71.1	75.4	79.0	74.4	78.4	73.5	77.9	14.2	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
74.4	134.8	76.6	73.3	129.8	75.9	72.3	125.4	75.6	39.8	79.1	38.8	78.5	38.0	78.2	79.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
28.5	25.7	23.4	DB	54.9	85.3	1.0	0.9	54.2	85.9	53.6	86.5	53.0	87.0	52.2	87.6
			WB	52.1	75.9	1.0	1.9	51.4	77.3	50.8	78.4	50.2	79.5	49.5	80.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	71.7	68.4	68.5	68.8	70.5	71.3	73.0	74.5	74.9	74.7	73.7	71.7	69.8
	DBStd	3.04	2.04	2.43	1.95	1.84	1.86	1.44	1.61	1.82	1.94	1.88	1.89	2.05
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	4	1	2	1	0	0	0	0	0	0	0	0	0
	CDD50	7912	571	518	582	615	661	690	758	773	742	736	651	615
	CDD65	2441	107	100	118	165	196	240	293	308	292	271	201	150
	CDH74	9014	199	191	179	376	615	962	1357	1604	1473	1192	581	285
	CDH80	711	6	7	1	6	22	49	116	194	169	110	23	8
	CDH80	711	6	7	1	6	22	49	116	194	169	110	23	8
(14) Wind	WSAvg	9.9	8.7	10.2	10.2	11.4	9.3	10.3	11.1	10.7	9.7	8.7	9.5	9.6
(15) Precipitation	PrecAvg	37.5	5.9	4.0	3.7	3.1	3.2	1.4	2.0	1.5	2.2	2.8	3.7	4.6
	PrecMax	66.5	16.9	13.7	12.8	14.3	10.9	6.5	7.4	4.1	8.9	11.5	16.5	10.4
	PrecMin	17.0	0.4	0.6	0.2	0.3	0.3	0.0	0.2	0.2	0.2	0.1	0.2	0.2
	PrecStd	12.5	5.0	3.4	3.0	3.1	2.9	1.4	1.6	1.0	2.2	2.6	3.8	3.2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	80.8	81.0	80.4	81.0	82.2	83.0	84.4	85.1	85.2	84.8	82.5	81.2
		MCWB	68.2	69.1	68.7	68.9	70.4	70.4	72.1	71.4	72.8	72.2	69.9	69.6
	2%	DB	78.1	78.4	77.7	79.0	80.5	81.5	82.6	83.4	83.2	82.6	80.4	78.6
		MCWB	67.5	68.0	67.5	68.4	69.7	70.0	71.5	71.6	71.9	71.5	69.3	68.4
	5%	DB	76.2	76.4	76.3	77.6	78.9	80.2	81.4	82.2	82.0	81.3	78.9	77.0
		MCWB	66.7	67.0	67.0	67.9	69.2	69.8	71.3	71.5	71.9	71.1	68.9	68.0
	10%	DB	74.7	74.6	74.9	76.3	77.4	78.9	80.2	81.0	80.9	79.9	77.4	75.4
		MCWB	66.2	66.3	66.5	67.5	68.5	69.5	71.1	71.3	71.6	70.7	68.4	67.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	70.8	72.3	70.6	71.8	73.1	74.4	75.8	76.9	76.7	76.1	72.8	73.0
		MCDB	74.7	75.2	76.6	75.5	77.1	77.0	78.8	79.8	79.7	79.2	77.6	76.0
	2%	WB	69.5	70.7	69.4	70.3	71.7	72.7	74.4	75.2	75.2	74.1	71.4	71.1
		MCDB	73.6	74.2	74.2	75.1	76.7	76.9	78.3	79.1	79.1	78.5	76.1	74.4
	5%	WB	68.7	69.3	68.6	69.3	70.7	71.4	73.4	74.0	74.2	73.1	70.5	70.0
		MCDB	73.2	73.3	73.6	74.8	76.1	77.1	77.9	78.3	78.5	78.0	75.6	73.7
	10%	WB	67.8	68.2	67.8	68.5	69.7	70.5	72.4	73.0	73.2	72.1	69.6	69.0
		MCDB	72.8	72.5	72.9	74.1	75.3	76.6	77.5	77.7	77.9	77.4	75.0	73.1
(35) Mean Daily Temperature Range	5% DB	MDBR	14.2	13.7	13.0	13.3	14.0	13.9	13.4	13.9	13.7	13.9	13.2	12.8
		MCDBR	16.1	16.1	14.7	14.5	15.2	15.1	14.3	14.8	14.6	15.3	14.7	14.5
		MCWBR	8.3	8.0	7.4	6.7	7.0	6.5	6.3	6.4	6.4	6.9	6.9	7.3
	5% WB	MCDBR	13.9	13.6	12.6	13.2	14.1	13.4	12.4	12.0	12.4	13.5	12.8	12.7
		MCWBR	7.7	7.6	7.0	6.7	6.8	6.5	6.1	6.1	6.2	6.6	6.6	7.4
(40) Clear-Sky Solar Irradiance	taub	0.333	0.338	0.347	0.362	0.359	0.347	0.346	0.346	0.339	0.346	0.343	0.331	
	taud	2.653	2.631	2.614	2.542	2.581	2.624	2.639	2.642	2.668	2.655	2.662	2.680	
	Ebn at Noon	301	305	304	298	296	297	297	299	302	298	295	298	
	Edh at Noon	27	29	31	34	32	30	30	30	29	28	27	26	
(44) All-Sky Solar Radiation	RadAvg	1442	1623	1826	2076	2203	2281	2255	2187	2042	1731	1465	1324	
	RadStd	65	103	175	120	118	61	49	87	56	47	36	107	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+162	+164

Nomenclature: See separate page